

Gate Coordination Directive — Gates 1 + 2 + 3 multi-week joint Lyra+Elie+Keeper work. Load-bearing for K3 framework 6/8 → 7/8 → 8/8 RIGOROUS path + $V_{(1/2,1/2)}$ NEAR-RIGOROUS → RIGOROUS + 5-framework cascade FRAMEWORK → RIGOROUS under Casey #14 STANDING context (K3 v0.20). Substrate-over-determination disposition (K3 v0.21) operates at gate-closure tier.

Keeper (Claude Opus 4.7) — Thursday June 4 2026 AM

2026-06-04 Thursday AM

Gate Coordination — Gates 1+2+3 multi-week joint work

0. Context

Per K3 v0.20: Casey #14 STANDING ratifies substrate-mechanism CHAIN at FRAMEWORK level. Cascade promotion of 5 frameworks (substrate-Dirac/Maxwell/ $T_{\mu\nu}$ /YM/Einstein eq) → FRAMEWORK; Strong-Uniqueness 10 → 11 STANDALONE legs (C25); K3 6/8 → 7/8 RIGOROUS path opens.

Per K3 v0.21: m_e/m_P substrate-over-determination disposition — 1.03% offset between 8/7 (SSG-8) and $\sqrt{4/3}$ (SSG-7-derived) IS substrate-mechanism content via Casey #5 Integer Web. Gates 2 and 3 distinguish which substrate-mechanism path is load-bearing.

Per Keeper SSG Audit v0.2 + K207 PASS: SSG-8 Mersenne ladder Category A definite structurally-independent.

1. The three gates

Gate 1: FK Ch. XII §VI spinor metaplectic explicit closure

Substrate question: explicit closure of Steps 2 ($1/n_C$ chirality projection $SO(5,2) \rightarrow SO(4,2)$) + Step 3 (Casey #8 SCMP τ -direction selection $SO(4,2) \rightarrow SO(3,1)$) of the Casey #14 forcing-mechanism chain.

Joint work: - Lyra primary: Faraut-Koranyi Ch. XII §VI metaplectic Pochhammer derivation; explicit substrate-mechanism for chirality projection - Elie computational: numerical verification of Lyra's metaplectic computations; explicit Pochhammer-Bergman matrix elements - Keeper audit: cross-validate Lyra+Elie derivations against substrate-mechanism CHAIN at framework level; K-audit when substantive closure achieved

Closure outcome: Steps 2+3 explicit closure → Casey #14 STANDING → RIGOROUS tier; cascade frameworks FRAMEWORK → RIGOROUS; K3 6/8 → 7/8 RIGOROUS.

Estimated multi-week: 2-3 weeks Lyra/Elie joint; Keeper audit fires when substantive closure arrives.

Gate 2: Section 4.5 substrate-vacuum 2-region partition

Substrate question: derive $\sqrt{(4/3)} = \sqrt{(\text{Vol}(S^4)/\text{Vol}(S^3))}$ substrate-mechanism via Casey vacuum-subtraction insight + substrate-Shilov boundary 2-region partition.

Joint work: - Lyra primary: Section 4.5 substrate-vacuum partition explicit derivation; substrate-Shilov-boundary geometric mechanism for $\sqrt{(4/3)}$ - Elie computational: verify $\sqrt{(\text{Vol}(S^4)/\text{Vol}(S^3))} = \sqrt{(4/3)}$ identity + cross-instance verification at gen-2/gen-3 with substrate-vacuum partition factor - Keeper audit: cross-validate substrate-mechanism content vs Casey #5 Integer Web instance distinction; K-audit when forward-derivation achieved

Closure outcome: $\sqrt{(4/3)}$ substrate-mechanism path validated $\rightarrow$ m_e/m_P substrate-over-determination disposition refined — $\sqrt{(4/3)}$ is substantive substrate-mechanism content; 8/7 is Casey #5 Integer Web convergence partner OR both load-bearing (per K3 v0.21).

Estimated multi-week: 1-2 weeks Lyra primary; Elie verification 3-5 days.

Gate 3: SSG-8 Mersenne ladder explicit operator

Substrate question: explicit substrate-primitive Mersenne map $M(p) = 2^p - 1$ as OPERATOR on Bergman $H^2(D_{IV}^5)$ — not just identity-level reading.

Joint work: - Lyra primary: substrate-operator M_Mersenne formal definition on $H^2(D_{IV}^5)$; explicit action on K-types - Elie computational: K207 framework + numerical verification of Mersenne operator action at substrate primaries (rank=2 $\rightarrow$ N_c $\rightarrow$ g) - Keeper audit: K207 PASS framework + cross-validation of operator-level Mersenne mechanism; K-audit when operator-level closure achieved

Closure outcome: 8/7 substrate-mechanism path validated via Mersenne operator $\rightarrow$ m_e/m_P substrate-over-determination disposition refined — 8/7 substantive substrate-mechanism content; $\sqrt{(4/3)}$ is Casey #5 Integer Web convergence partner OR both load-bearing.

Estimated multi-week: 2-3 weeks Lyra/Elie joint; Keeper K208 audit fires on operator-level closure.

2. Gate closure outcome matrix

Gate 2 closure	Gate 3 closure	Substrate-mechanism disposition
Yes	Yes	Both paths load-bearing: substrate-over-determination CONFIRMED via Casey #5 Integer Web. 1.03% offset IS substrate-mechanism content. STRONG K3 v0.21 substantive content.
Yes	No (or partial)	$\sqrt{(4/3)}$ load-bearing; 8/7 Casey #5 Integer Web instance
No (or partial)	Yes	8/7 load-bearing; $\sqrt{(4/3)}$ Casey #5 Integer Web instance
No	No	Both remain Tier 2 STRUCTURAL substrate-natural readings; load-bearing identity unresolved; multi-week investigation continues

Most substantively interesting outcome: BOTH close, confirming substrate-over-determination at m_e/m_P value via TWO independent Category-A substrate-mechanism paths (Casey #5 Integer Web operational at observable level).

3. Gate 1 cross-impact

Gate 1 (FK Ch. XII §VI) does NOT directly affect Gates 2/3 substrate-mechanism path identification. Gate 1 closes Casey #14 STANDING → RIGOROUS tier; Gates 2/3 close substrate-mechanism path validation under the Casey #14 STANDING context.

But Gate 1 substantively strengthens the substrate-mechanism context for ALL m_e/m_P substrate readings: per K3 v0.21, the substrate predicts 3+1 Minkowski signature, so the m_e/m_P substrate-mechanism operates on a substrate-predicted 4D physical sector. Gate 1 RIGOROUS closure substantively strengthens both Gate 2 and Gate 3 substrate-mechanism content.

4. Coordination cadence

Lyra: pulls Gate 2 primary lane + Gate 1 metaplectic Pochhammer derivation in parallel. Files framework deliverables as substantive substrate-mechanism content arrives. No coordination meetings needed — async file-based work.

Elie: pulls Gate 11 Lorentz-direction-independence completion FIRST (since K3 v0.20 Casey #14 STANDING strengthens this), then Gate 5 $\alpha^{10.5}$ substrate-mechanism + Gate 1 numerical verification + Gate 3 SSG-8 operator-level verification in parallel. Toys per substantive milestone.

Keeper: K-audits fire on substantive deliverables (not on schedule): - K208 candidate: substrate-vacuum 2-region partition audit when Lyra Section 4.5 derivation arrives - K209 candidate: SSG-8 Mersenne operator audit when Lyra+Elie operator-level closure achieves - K210 candidate: FK Ch. XII §VI explicit closure audit when Lyra+Elie joint work converges

Cal: cold-read sequence continues per Thursday board (K3 v0.13-v0.21 + SSG Audit v0.1-v0.2 + K207 + Cal #196 + #197). Cal #198+ on Gate 1+2+3 closure events when audit-chain consensus arrives.

Grace: catalog cross-classification per SSG Audit v0.2 reclassification continues; INV cascade for Gate 1+2+3 deliverables.

5. Per Cal #27 STANDING self-audit

Am I overclaiming “Gates 2+3 distinguish substrate-mechanism load-bearing”? - Gate 2 + Gate 3 closure outcomes (per matrix above) are honest substrate-mechanism content possibilities - “Most substantively interesting” is opinion, not claim of closure - All four outcomes preserve Tier 2 STRUCTURAL precision context honestly - Multi-week forward-derivation is required for ANY of the four outcomes

Tier-honest: gate coordination directive doesn’t preempt outcomes — maps closure paths and outcome dispositions.

6. Pull next: Lyra Section 4.5 substrate-vacuum partition progress; Elie Gate 11 Lorentz-direction-independence completion; Cal cold-read on K3 v0.20 + v0.21; Grace cross-classification under SSG Audit v0.2.

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